

HARPREET SINGH

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SUMMARY

Electromechanical Engineering Technology (Automation) graduate from Seneca Polytechnic with hands-on experience across the full controls stack: Allen-Bradley and Siemens PLC programming, HMI development, KUKA robotics integration, EPLAN panel design, commissioning, and live fault diagnosis. Trained in BACnet building automation, industrial networks, pneumatics, and detailed technical documentation. Toronto-based, valid G-class driver's license with personal vehicle, open to relocation across Canada, comfortable with shift work and travel. Working toward OACETT C.E.T. designation.

TECHNICAL SKILLS

PLC and Controls: Allen-Bradley Studio 5000, CompactLogix, Siemens TIA Portal, CLICK PLC, Ladder Logic, Structured Text, PID Control and Tuning, Analog I/O, VFD and Motion Control

HMI and Robotics: HMI Development with VB Scripting, Animated Operator Screens, Alarm Management, Node-RED Dashboards, KUKA KRL Programming (Pendant and WorkVisual), PLC-Robot Integration, Safety Circuits, NX MCD Digital Twin

Electrical and Panel Building: EPLAN Electrical Schematics, Control Panel Wiring, 600/480VAC, 120VAC, 24VDC Power Distribution, Loop Diagrams, Sensors, Servos, Actuators, Motor Drives, Pneumatics, Multimeter Diagnostics

Commissioning and Maintenance: System Startup, I/O Mapping, Wiring Checkout, Sensor Calibration, Point-to-Point Verification, Fault Diagnosis, Root Cause Analysis, Preventive Maintenance, Component Replacement, Detailed Service Records

Industrial Networks and Building Automation: EtherNet/IP, Modbus RTU/TCP, OPC-UA, MQTT, BACnet/IP, BACnet MS/TP, HVAC Controls (AHU, VAV, DDC), Kantech Access Control

CAD and Manufacturing: EPLAN, SolidWorks, NX CAD/CAM, Mastercam, HAAS Mill Operation, CNC Setup

HANDS-ON TECHNICAL EXPERIENCE

Multi-Station Automated Production System (Capstone)

2025

- Designed, wired, programmed, and commissioned a multi-station automated system using ladder logic on a CLICK PLC, photoeye and proximity sensors triggering pneumatic diverter logic, with the panel built from EPLAN schematics and full I/O documentation.
- Performed live commissioning including I/O checks, point-to-point verification, sensor calibration, and root cause analysis on real faults including stuck cylinders, miswired sensor inputs, and incorrect actuator timing.
- Built a Node-RED and MQTT dashboard tracking throughput, equipment status, and fault codes for real-time monitoring and continuous improvement analysis.

Allen-Bradley and Siemens PLC, HMI, and PID Programming

2024

- Programmed CompactLogix in Studio 5000 and Siemens TIA Portal with ladder logic, structured text, PID loops, analog I/O scaling, and animated HMI screens; configured inter-PLC communication.
- Built a closed-loop PID temperature control system with RTD sensor input and heater plus fan output, calibrated sensors, and tuned PID parameters for stable setpoint tracking.

KUKA Robot and PLC Integration

2025

- Programmed KUKA industrial robot in KRL via teach pendant and WorkVisual, integrated with PLC for synchronized pick-and-place, configured safety circuits and I/O handshaking, validated motion paths in NX MCD digital twin.

EDUCATION AND CERTIFICATIONS

Advanced Diploma, Electromechanical Engineering Technology (Automation)

May 2023 to December 2025

Seneca Polytechnic, Toronto, ON | GPA 3.2/4.0

Siemens TIA Portal, Control Systems Fundamentals, Industrial IoT, Mechatronics Level 1 | KUKA Robot Programming I, Robot Applications | EPLAN Certified Student | BACnet Basics, BACnet Device Profiles | HAAS Mill Operator | WHMIS | Working toward OACETT C.E.T. designation

WORK EXPERIENCE

Building Systems and Security Monitor | Paragon Security, Toronto, ON

May 2024 to Present

- Operate Kantech access control and building safety systems across commercial facilities on rotating 12-hour shifts including nights and weekends, monitoring alarms and system events in real time.
- Maintain detailed shift handover logs, incident reports, and service records, coordinating with facility staff, contractors, and external vendors to resolve issues under time pressure.

Assistant Manager | Subway, Toronto, ON

June 2023 to May 2024

- Supervised daily operations, inventory, and team training; maintained accurate records across opening and closing shifts.